

Smart Home Automation

Wireless Appliance Control via Bluetooth and Active-Low Relay

1. Project Overview

The objective of this project is to design and implement a cost-effective, wireless home automation system. By integrating an Arduino microcontroller with a Bluetooth module and an active-low relay board, users can seamlessly control household AC appliances (such as lights, fans, and sockets) directly from their smartphone. This system bridges the gap between digital mobile interfaces and physical electrical hardware.

2. Hardware Components

The system is built using readily available electronic components that ensure reliability and ease of programming.

Arduino Uno (Microcontroller)

Acts as the brain of the system. It receives wireless data, processes the logical commands, and sends control signals to the relay module to toggle appliances.

HC-05 / HC-06 Bluetooth Module

This module provides a wireless Serial Interface (UART) between the smartphone and the Arduino. It receives commands transmitted from a mobile Bluetooth terminal app and forwards them to the Arduino's RX pin.

Active-Low Relay Module

A relay acts as an electrically operated switch, allowing the low-voltage Arduino (5V) to safely control high-voltage AC mains (220V/110V). **Active-Low operation** means that the relay triggers (turns ON) when it receives a LOW (0V) signal from the Arduino, and it deactivates (turns OFF) when it receives a HIGH (5V) signal.

3. System Architecture

The data flow of the system is linear and highly responsive. When a user taps a button on their mobile app, a specific character is transmitted over Bluetooth.



4. Working Principle

The working mechanism centers around Serial Communication.

- **Initialization:** Upon booting, the Arduino initializes the relay control pins to HIGH to ensure all appliances are turned OFF by default (due to the active-low nature of the relay).
- **Transmission:** The user opens a Bluetooth terminal app on their mobile device, pairs with the HC-05, and sends specific characters (e.g., '1' for ON, '0' for OFF).
- **Execution:** The Arduino reads the incoming serial data. If it reads a '1', it executes `digitalWrite(RELAY_PIN, LOW);`, activating the relay electromagnet, closing the AC circuit, and turning the appliance ON.
- **Reversal:** If it reads a '0', it executes `digitalWrite(RELAY_PIN, HIGH);`, deactivating the relay and breaking the AC circuit.

Important Note on Active-Low Relays: Because the relay activates on a LOW signal, the logical states in the Arduino code are inverted compared to standard LEDs. LOW = Device ON, HIGH = Device OFF.

5. Firmware / Arduino Code Snippet

The core logic has been structured sequentially. The code leverages custom functions (modularity) and standard C++ switch-case logic to process commands linearly, making it highly readable and robust for future scaling.

```

// --- Pin Definitions & Constants ---
const int RELAY_PIN = 8;
const int BAUD_RATE = 9600;

// --- Setup Routine ---
void setup() {
    // 1. Initialize Serial Communication for HC-05
    Serial.begin(BAUD_RATE);

    // 2. Configure Pin Modes
    pinMode(RELAY_PIN, OUTPUT);

    // 3. Set Initial State (Active-Low: HIGH = Appliance OFF)
    digitalWrite(RELAY_PIN, HIGH);

    Serial.println("System Initialized. Awaiting input...");
}

// --- Main Loop ---
void loop() {
    // Continuously monitor incoming Bluetooth data
    if (Serial.available() > 0) {
        char command = Serial.read();
        executeCommand(command);
    }
}

// --- Modular Command Execution ---
void executeCommand(char cmd) {
    switch(cmd) {
        case '1':
            digitalWrite(RELAY_PIN, LOW); // Pull LOW to turn ON
            Serial.println("Status: Appliance ON");
            break;
        case '0':
            digitalWrite(RELAY_PIN, HIGH); // Push HIGH to turn OFF
            Serial.println("Status: Appliance OFF");
            break;
        default:
            // Ignore any invalid characters
            break;
    }
}

```

6. Applications and Advantages

- **Convenience & Comfort:** Users can control room lighting or fans from their bed or couch without physically interacting with switches.
- **Accessibility:** Highly beneficial for elderly or differently-abled individuals who may find reaching traditional wall switches difficult.
- **Scalability:** The system can be easily expanded by using a 4-channel or 8-channel relay board to control multiple appliances simultaneously from a single Arduino.

7. Conclusion

This project successfully demonstrates the integration of mobile technology with hardware switching. Using an Arduino and a Bluetooth module provides a robust, localized wireless control system. The application of an active-low relay ensures safe isolation between the high-voltage AC mains and the low-voltage DC microcontroller logic, resulting in a safe, functional, and highly practical home automation solution.